159102 workshop – Week 9

1. What is the resolution of the screen you are using? You can usually find this by looking at Settings and Display or by using this: https://bestfirms.com/what-is-my-screen-resolution/
(*if you are unable to do this, don't worry about it*)

2. Find a useful website to help you answer the following. You could try using: https://www.rapidtables.com/web/color/RGB_Color.html

(a) RGB values for a nice shade of yellow

(b) RGB values for a nice shade of purple

(c) RGB values for a nice shade of brown

No real answer for these – it is entirely up to you.

3. Cyan is a colour that is talked about a lot. You should be aware what cyan looks like. Write a program to display a cyan rectangle. The RGB values are 0 255 255.
These are the RGB values for "standard" cyan. Then people get onto "dark cyan", etc...

4. What colour do you get if the three RGB values are the same? E.g. 150, 150, 150. Leave out the small numbers because we know that 0, 0, 0 is black. And leave out the large numbers because we know that 255, 255, 255 is white. Look somewhere in between. Note that this colour can appear in many shades.

5. Assume you want to write a program to create an image in a file called smallimage.ppm. Your image is going to be 400 pixels down the screen and 250 pixels across the screen. Write down the first four lines that will appear in the file called smallimage.ppm.

6.** Find (or invent) an interesting flag. Modify program 3 in the notes to create the image of this flag. WARNING – some flags are very challenging. The flags of NZ or Australia are very difficult to get right and the flag of Brazil is a total nightmare.

7. What does the following system command do: F:
(*if you are unable to do this on your computer, don't worry about it*)

8. The following program is meant to create the image of a green rectangle. However there are five (5) errors in the program. Identify each error and state what needs to be done to correct it.

```
#include <iostream>
#include <fstream>
using namespace std;

class pixel_class {
private:
  int red, green, blue;
public:
```

```
  void loaddata(int v1, int v2, int v3);
  void datatofile(fstream ppmfile);
  int getR() { return red; }
  int getG() { return green; }
  int getB() { return blue; }
};

void createpicture();

pixel_class picture[500][300];

int main() {
  int x, y;
  fstream myfile;
  createpicture();
  filename = "myImage.ppm";
  myfile.open(filename.c_str(), fstream::out);
  myfile << "# " << filename << endl;
  myfile << 250 << " " << 100 << endl;
  myfile << 256 << endl;
  for (y = 0; y < 300; y++) {
    for (x = 0; x < 500; x++) {
      picture[x][y].datatofile(myfile);
    }
    myfile << endl;
  }
  myfile.close();
}

void createpicture() {
  int x, y;
  for (y = 0; y < 300; y++) {
    for (x = 0; x < 500; x++) {
      picture[x][y].loaddata(255, 153, 0);
    }
  }
}

//--------------- methods for the pixel_class ------------
void pixel_class::loaddata(int v1, int v2, int v3) {
  red = v1;
  green = v2;
  blue = v3;
}

void pixel_class::datatofile(fstream & ppmfile) {
  // write the data for one pixel to the ppm file
  ppmfile << red << " " << green;
  ppmfile << " " << blue << "  ";
}
```